

Part I

Some Language and Notation of Mathematics

1 August 31st: Sets

Definition 1.0.1

A **set** is a collection of elements.

Intuitively, a set can be thought of as a bag—our set—which may or may not contain stuff—our elements. Sets are the most fundamental objects in mathematics and many important concepts, such as functions and spaces, can be defined using sets.

If we want to say that an element x belongs to some set S , we write

$$x \in S$$

with the “ $\in$ ” symbol meaning “an element of” (so $x \in S$ literally means x is an element of S).

When defining a set, we use curly braces “ $\{\}$ ” to enclose the elements inside of it.

Example 1.0.2. The following are some basic examples of sets:

1. $A := \{0, 3, \heartsuit\}$ defines a set, named A , to the elements 0, 3, and $\heartsuit$ (elements do not necessarily need to be numbers, indeed another set can also be an element!)
2. $B := \{\text{all odd integers}\}$ defines a set, named B , to all odd integers (e.g. $-1 \in B$ and $127 \in B$).
3. $C := \{n : n \text{ is an integer satisfying } n^3 = n\}$ (the colon “ $:$ ” in between the curly braces stands for “such that,” so C is the set of all integers n such that $n^3 = n$ (equivalently, $C := \{-1, 0, 1\}$))
4. The standard sets:

$\mathbb{N} := \{0^1, 1, 2, 3, \dots\}$ (the set of natural numbers)

$\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}$ (the set of integers)

$\mathbb{Q} := \left\{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\right\}$ (the set of rational numbers)

$\mathbb{R} :=$ the set of real numbers²

$\mathbb{C} := \{a + bi : a, b \in \mathbb{R}, i = \sqrt{-1}\}$ (the set of complex numbers)

Remark 1.0.3. Order and repetition in sets do not matter. Indeed, the set $A := \{0, 3, \heartsuit\}$ is equal to the sets $A_1 := \{3, \heartsuit, 0\}$ and $A_2 := \{0, 3, \heartsuit, 0\}$.

¹Some sources omit 0 from the set of natural numbers, and define $\mathbb{N} := \{1, 2, 3, \dots\}$. However in this class, we will always assume $0 \in \mathbb{N}$, and define $\mathbb{N}^+ := \{1, 2, 3, \dots\}$.

²The set of real numbers is surprisingly difficult to define. Take MATH255 to learn how to rigorously define $\mathbb{R}$.

Definition 1.0.4

Two sets A and B are said to be equal if and only if they have the same elements. Symbolically, we write $A = B \iff \forall x, x \in A \text{ when } x \in B$.

In the above theorem, we introduced two key symbols often used in mathematics. The first, the “ $\iff$ ” symbol, stands for “if and only if.” In mathematics, this means that the statement on the left is equivalent to the statement on the right. The second symbol, the “ $\forall$ ”, stands for “for all” or “for every” and is called the **universal quantifier**. Symbolic notation is often eschewed in formal writing but is used very often in more casual mathematical writing.

Remark 1.0.5. By Definition 1.0.3, we see that the set A in Remark 1.0.2 is equal to the sets A_1 and A_2 .

Definition 1.0.6

Let A and B be arbitrary sets. We say that A is a **subset of** B , or A is **contained in** B , if for every $x \in A$, $x \in B$. In this case we write, $A \subseteq B$, where “ $\subseteq$ ” is the subset symbol³.

Remark 1.0.7. Note that $A = B$ holds when $A \subseteq B$ and $B \subseteq A$ both hold. This gives us a very useful way to prove if two sets are equal (we simply prove that one set is the subset of the other and vice versa).

Definition 1.0.8

The set with no elements is called the **empty set**, and is denoted $\emptyset := \{\}$.

Intuitively, the empty set can be thought of as a bag with nothing inside of it. The empty set is *not* nothing, but it contains nothing.

Example 1.0.9. We conclude with some examples of sets and some of their properties:

1. For every set A , $\emptyset \subseteq A$.
2. $\emptyset \neq \{\emptyset\}$, indeed we see that $\emptyset \subseteq \{\emptyset\}$ but $\{\emptyset\} \not\subseteq \emptyset$.
3. $\{\emptyset, \{\emptyset\}\} \neq \{\{\emptyset\}\}$
4. In set theory, we write the numbers $0 := \emptyset$, $1 := \{\emptyset\}$, $2 := \{\emptyset, 1\}$, and so on.

³Many textbooks use the symbol “ $\subset$ ” to denote a subset instead of $\subseteq$, i.e. if A is a subset of B , then $A \subset B$ is written. In these notes, I will use the symbol $\subset$ only when one set is a *proper subset* of the other, in the same way that $<$ defines a strict inequality.

2 September 2nd: Set Operations

2.1 Basic Set Operations

Definition 2.1.1

Let A, B be two sets. We define:

- Their **union**, denoted $A \cup B$, as the set $\{x: x \in A \text{ or } x \in B\}$
- Their **intersection**, denoted $A \cap B$, as the set $\{x: x \in A \text{ and } x \in B\}$
- Their **difference**, denoted $A \setminus B$, as the set $\{x: x \in A \text{ and } x \notin B\}$.

If A is the ambient set (say for instance, $A = \mathbb{Z}$, then when $B \subseteq A$ we sometimes write $B^c := A \setminus B$ and call it the **complement** of B relative to A .

Example 2.1.2. Within $\mathbb{Z}$, define $A := \{n \in \mathbb{Z}: n \text{ is even}\}$. We have that the complement of A , $A^c := \{\text{set of odd integers}\}$.

Definition 2.1.3

For two elements a and b (a and b are not necessarily sets), we define their **ordered pair**, denoted (a, b) to be the tuple satisfying the property $(a, b) \neq (b, a)$ whenever $a \neq b$.

We will use this notion of ordered pairs to help us with the following important definition:

Definition 2.1.4

Let A, B be two sets. We define their **cartesian product**, denoted $A \times B$, as the set of ordered pairs $\{(a, b): a \in A \text{ and } b \in B\}$.

A notable cartesian product is the Euclidean plane $\mathbb{R}^2 := \mathbb{R} \times \mathbb{R}$. You have already seen this plane defined as the xy -plane, where each ordered pair (x, y) represents the x - and y -coordinates of a point on the plane.

Example 2.1.5. Let $A := \{0, 1, 2\}$ and $B := \{a, b\}$. Then their cartesian product $A \times B$ is the set $\{(0, a), (0, b), (1, a), (1, b), (2, a), (2, b)\}$.

Remark 2.1.6. In general, if A has n elements and B has m elements, then their cartesian product $A \times B$ has $n \cdot m$ elements.

Definition 2.1.7

Let A be a set. We say the **power set of A** , denoted 2^A or $\mathcal{P}(A)$, is the set of all subsets of A , which includes A and the empty set $\emptyset$.

Example 2.1.8. Let $A := \{0, 1, 2\}$, then its power set $\mathcal{P}(A)$ is:

$$\mathcal{P}(A) := \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$$

Remark 2.1.9. For each element in a nonempty set, we can either choose to include or not include

it in a subset of A . This means that for a set A with n elements, there are 2^n elements in its power set.

2.2 Algebra of Set Operations

We will show our first proposition (a statement which we will prove) here.

Proposition 2.2.10

Let A , B , and C be arbitrary sets. Then the following properties hold:

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
3. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
4. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$
5. $(A^c)^c = A$

Proof. Each point in this proposition claims that two sets are equal to each other. By Remark 1.0.6., we will show that each condition holds by showing the two sides of the equality are subsets of each other.

Let us start by proving (1). We first want to show that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$. How do we do this? Let us start by choosing an arbitrary element $x \in A \cap (B \cup C)$. If the subset condition holds, then x should also be an element in $(A \cap B) \cup (A \cap C)$. Hence, we want to prove that if $x \in A \cap (B \cup C)$, then $x \in (A \cap B) \cup (A \cap C)$.

We start by letting x be an arbitrary element in $A \cap (B \cup C)$. By definition of union and intersection, then $x \in A$ and $x \in B \cup C$. However we do not know if $x \in B$ or if $x \in C$. To account for this, we will split out proof into two cases.

Firstly, we suppose $x \in B$. Then, $x \in A \cap B$, which immediately implies that $x \in (A \cap B) \cup (A \cap C)$. Secondly, we suppose $x \in C$. Then $x \in A \cap C$, which again implies that $x \in (A \cap B) \cup (A \cap C)$. Hence we have shown $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and the first half of the proof is complete.

We will now prove the converse condition holds, i.e. $A \cap (B \cup C) \supseteq (A \cap B) \cup (A \cap C)$. Let $x \in (A \cap B) \cup (A \cap C)$. By definition, we have that $x \in A \cap B$ or $x \in A \cap C$, but we do not know which. So let us again use cases. Firstly, if $x \in A \cap B$, then $x \in A$ and $x \in B$, so $x \in A \cap (B \cup C)$. Secondly, if $x \in A \cap C$, then $x \in A$ and $x \in C$, so $x \in A \cap (B \cup C)$. Hence we have shown $A \cap (B \cup C) \supseteq (A \cap B) \cup (A \cap C)$, so condition (1) holds.

The proof for (1) was very wordy for the sake of exposition. We prove the following conditions much more tersely.

For (2), let $x \in A \cup (B \cap C)$. By definition, $x \in A$ or $x \in B \cap C$. In the case $x \in A$, then $x \in (A \cap B) \cup (A \cap C)$. In the case $x \in B \cap C$, then $x \in B$ and $x \in C$, which implies $x \in (A \cap B) \cup (A \cap C)$. Hence $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

Conversely, let $x \in (A \cup B) \cap (A \cup C)$. By definition, $x \in (A \cup B)$ and $x \in (A \cup C)$, which implies $x \in A$ or $x \in (B \cap C)$, which by definition implies that $x \in A \cup (B \cap C)$. Hence $x \in (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$ and (2) holds.

For (3), let $x \in A \setminus (B \cup C)$. By definition, $x \in A$ and $x \notin (B \cup C)$. Since $x \notin (B \cup C)$, $x \notin B$ and $x \notin C$ ³. Hence $x \in A$ and $x \notin B$, and $x \in A$ and $x \notin C$ all hold true, which means $x \in (A \setminus B) \cap (A \setminus C)$. Hence $A \setminus (B \cup C) \subseteq (A \setminus B) \cap (A \setminus C)$.

Conversely, let $x \in (A \setminus B) \cap (A \setminus C)$. By definition, $x \in A \setminus B$ and $x \in A \setminus C$, which means $(x \in A \text{ and } x \notin B)$ and $(x \in A \text{ and } x \notin C)$. Rearranging this gives $x \in A$ and $(x \notin B \text{ and } x \notin C)$. Since $x \notin B$ and $x \notin C$, it follows that $x \notin (B \cup C)$. Thus, $x \in A$ and $x \notin (B \cup C)$, which implies $x \in A \setminus (B \cup C)$. Hence $(A \setminus B) \cap (A \setminus C) \subseteq A \setminus (B \cup C)$, and (3) holds.

For (4), let $x \in A \setminus (B \cap C)$. By definition, $x \in A$ and $x \notin (B \cap C)$. Since $x \notin (B \cap C)$, we have $x \notin B$ or $x \notin C$. This means $(x \in A \text{ and } x \notin B)$ or $(x \in A \text{ and } x \notin C)$, which by definition implies that $x \in A \setminus B$ or $x \in A \setminus C$. Hence $x \in (A \setminus B) \cup (A \setminus C)$, and $A \setminus (B \cap C) \subseteq (A \setminus B) \cup (A \setminus C)$.

Conversely, let $x \in (A \setminus B) \cup (A \setminus C)$. By definition, $x \in A \setminus B$ or $x \in A \setminus C$, which means $(x \in A \text{ and } x \notin B)$ or $(x \in A \text{ and } x \notin C)$. Factoring out $x \in A$ gives $x \in A$ and $(x \notin B \text{ or } x \notin C)$. Since $x \notin B$ or $x \notin C$, it follows that $x \notin (B \cap C)$. Thus $x \in A$ and $x \notin (B \cap C)$, so $x \in A \setminus (B \cap C)$. Hence $(A \setminus B) \cup (A \setminus C) \subseteq A \setminus (B \cap C)$, and (4) holds.

Lastly, (5) is so simple to prove that we do not even need to show that both sides of the equality are subsets each other. Instead, by the definition of the complement, we have:

$$A^c := \{x : x \notin A\}$$

Hence the complement of A^c is, by definition, the set of all elements that are not in the complement of A , or:

$$(A^c)^c := \{x : x \in A\}$$

which is equal to A .

We have successfully proved all five statements. To indicate that a proof is finished, one usually puts a square on the right side of the page, as is seen here: ■

Proposition 2.2.11 (De Morgan's Laws)

Let A be a set, and $B \subseteq A$ and $C \subseteq A$. Then the following properties hold:

1. $(B \cap C)^c = B^c \cup C^c$
2. $(B \cup C)^c = B^c \cap C^c$

Proof. For (1), let $x \in (B \cap C)^c$. By definition of the complement, $x \in A$ and $x \notin B \cap C$. Since $x \notin B \cap C$, it is not the case that both $x \in B$ and $x \in C$, which means $x \notin B$ or $x \notin C$. If $x \notin B$, then $x \in B^c$; if $x \notin C$, then $x \in C^c$. In either case, $x \in B^c \cup C^c$. Hence $(B \cap C)^c \subseteq B^c \cup C^c$.

Conversely, let $x \in B^c \cup C^c$. By definition, $x \in B^c$ or $x \in C^c$, which means $x \notin B$ or $x \notin C$. This is equivalent to saying $x \notin (B \cap C)$. Since $x \in A$ (given that $B, C \subseteq A$), it follows that $x \in (B \cap C)^c$. Thus $B^c \cup C^c \subseteq (B \cap C)^c$, and condition (1) holds.

For (2), let $x \in (B \cup C)^c$. By definition, $x \in A$ and $x \notin (B \cup C)$. Since x is not in the union $B \cup C$, we must have $x \notin B$ and $x \notin C$. This means $x \in B^c$ and $x \in C^c$, which implies $x \in B^c \cap C^c$. Hence $(B \cup C)^c \subseteq B^c \cap C^c$.

³We will prove this identity in Proposition 2.2.11, but for now a Venn diagram illustrates this point well.

Conversely, let $x \in B^c \cap C^c$. By definition, $x \in B^c$ and $x \in C^c$, meaning $x \notin B$ and $x \notin C$. This implies that $x \notin (B \cup C)$. Since $x \in A$, we conclude that $x \in (B \cup C)^c$. Thus $B^c \cap C^c \subseteq (B \cup C)^c$, and condition (2) holds. ■

3 September 4th: Proof by Contradiction, Contrapositive, and Induction

3.1 Proof by Contradiction

When we are asked to prove a statement P , we can prove P by assuming that P is false (i.e. $\neg P$ —the “ $\neg$ ” symbol means “not”—is true) and then showing that $\neg P$ cannot be true. This is called a **proof by contradiction**.

Proposition 3.1.1

$$\sqrt{2} \notin \mathbb{Q}.$$

Proof. Let us assume, by contradiction, that the negation of the proposition is true, i.e. we assume that $\sqrt{2} \in \mathbb{Q}$. Then, we can write $\sqrt{2} = p/q$, where p and q are two integers that are irreducible. Then $2 = p^2/q^2$ and $2q^2 = p^2$. Since p^2 is the product of 2 and q^2 , then p^2 is even, which immediately implies p is even. Since p is even, it can be written as $p = 2k$ where $k \in \mathbb{Z}$. Squaring both sides yields $p^2 = 4k^2$. Let $2q^2 = 4k^2$ and so $q^2 = 2k^2$, so q^2 is even which implies q^2 is also even. But we assumed that p and q are irreducible. Contradiction. ■

3.2 Proof by Contrapositive

Suppose we are trying to prove a statement $P \implies Q$, i.e. if P is true, then Q is true as well. This statement is logically equivalent to

$$\neg Q \implies \neg P$$

which is the **contrapositive** of $P \implies Q$. It is often easier to prove the contrapositive of the statement rather than proving the statement directly.

Lemma 3.2.2

Let $a, b \in \mathbb{Z}$. If ab is even, then a or b are even.

Proof. Let us prove the contrapositive of the statement: if both a and b are odd, then ab is odd. Assume a and b are even, then $a := 2k+1$ and $b := 2l+1$ for $k, l \in \mathbb{Z}$. We write $ab = (2k+1)(2l+1) = 4kl + 2k + 2l + 1 = 2(2kl + k + l) + 1$. Let $m := 2kl + k + l$, so $ab = 2m + 1$ and so ab is odd. ■

3.3 Proof by Induction

A key property of the natural numbers is that they are well ordered, namely every nonempty subset $S \subseteq \mathbb{N}$ has a unique least element n_0 (i.e. every element in S is less than or equal to n_0).

This property is known as the **axiom of well-orderedness**.

Theorem 3.3.3 (Principle of Induction)

Let $P \subseteq \mathbb{N}$. Suppose the following statements are true:

1. The base case: for $n_0 \in \mathbb{N}$, $n_0 \in P$.
2. The inductive step: for all $n \in \mathbb{N}$, if $n \in P$, then $n + 1 \in P$.

Then $P \supseteq \{n \in \mathbb{N} : n \geq n_0\}$.

Proof. By contradiction, suppose that $P \not\supseteq \{n \in \mathbb{N} : n \geq n_0\}$. Then the set $S := \{n \in \mathbb{N} : n \geq 0 \text{ and } n \notin P\} \neq \emptyset$. Let s_0 be the least element in S by the well-ordered axiom. We have that $s_0 \neq n_0$ since $n_0 \in P$. Thus $s_0 > n_0$ and $s_0 - 1 \geq n_0$. Since s_0 is a least element of S , $s_0 - 1 \notin S$ hence $s_0 - 1 \in P$. But by condition (2) $s_0 \in P$. Contradiction. ■

Example 3.3.4. We claim, for any $n \in \mathbb{N}$:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Proof. We will proceed by induction. We first consider the base case $n = 0$:

$$0 = \frac{0(1)}{2} = 0$$

Now, let us assume that the condition holds for every $n \in \mathbb{N}$. We want to show the condition holds for $n + 1$. We now want to prove:

$$1 + 2 + \cdots + n + (n + 1) = \frac{(n + 1)(n + 2)}{2}$$

By the inductive hypothesis, let us write $1 + 2 + \cdots + n = n(n + 1)/2$. By simple manipulations, we see that:

$$\begin{aligned} \frac{n(n+1)}{2} + (n+1) &= \frac{(n+1)(n+2)}{2} \\ \frac{n(n+1) + 2(n+1)}{2} &= \frac{(n+1)(n+2)}{2} \\ \frac{(n+1)(n+2)}{2} &= \frac{(n+1)(n+2)}{2} \end{aligned}$$

■

4 September 9th: Complete Induction, Pigeonhole Principle, Functions

4.1 Complete Induction

Theorem 4.1.1 (Complete Induction)

Let $n_0 \in \mathbb{N}$ and $P \subseteq \mathbb{N}$ satisfying that for each $n \geq n_0$, if $\{n_0, n_0 + 1, \dots, n - 1\} \subseteq P$, then $n \in P$. Then $P \supseteq \{n_0, n_0 + 1, \dots\} = \{n \in \mathbb{N} : n \geq n_0\}$.

Complete induction is often known as "**strong induction**", while Theorem 3.3.3 is often known as "**weak induction**".

Proof. Left as an exercise. ■

4.2 The Pigeonhole Principle

The Pigeonhole Principle is a powerful consequence of the well-ordered axiom which states that if there are n pigeons and m pigeonholes, where $n \geq m$, then at least one pigeonhole has two or more pigeons in it.

Proposition 4.2.2

Let $a_1, a_2, \dots, a_5 \in \mathbb{Z}$. If we divide all integers by 4, then there are two among them with the same remainder.

Proof. The possible remainders are $\{0, 1, 2, 3\}$. Since there are 5 integers that can have 4 different remainders, we invoke the pigeonhole principle to show that at least two among $a_1, \dots, a_5$ with the same remainder. ■

Proposition 4.2.3

For any group of greater than 2 people, there are two different people who have the same number of friends.

Proof. Left as an exercise. ■

4.3 Functions

Intuitively, we can think of functions as mapping elements from one set to a unique element in another. More formally:

Definition 4.3.4

A **function** from a set X to a set Y , denoted $f: X \rightarrow Y$, is a subset $f \subseteq X \times Y$ satisfying that for each $x \in X$, there exists a unique $y \in Y$ such that the ordered pair $(x, y) \in f$. If this is the case we write $f(x) = y$. The set X is called the **domain** of f while the set Y is called the **codomain** of f .

Example 4.3.5. The simplest function is called the **identity function**, which simply sends each element of a set X to itself. It can be written as $1_A: A \rightarrow A$, where $1_A(x) = x$ for all $x \in A$.

Definition 4.3.6

Let $f: X \rightarrow Y$ be a function. The **graph** of a f , often denoted Γ_f or simply Γ , is defined by set $\{(x, f(x)) \in X \times Y: f(x) = y\}$.

Example 4.3.7. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) := x^2$. Consider as well the function $g: \mathbb{R} \rightarrow [0, \infty)$ given by $g(x) := x^2$. We have that $\Gamma_f = \Gamma_g$, but f and g are technically two different functions. However, $f \equiv g$.

Definition 4.3.8

For a function $f: X \rightarrow Y$ and $A \subseteq X$, $B \subseteq Y$:

1. For $a \in A$, we define the **image of A under f** , denoted $f(A) := \{f(a) : a \in A\}$.
2. For $x \in X$, we define the **preimage of B** , denoted $f^{-1}(B) := \{x \in X : f(x) \in B\}$.

Remark 4.3.9. For $f: X \rightarrow Y$, $A \subseteq X$ and $B \subseteq Y$, $f^{-1}(f(A)) \supseteq A$, but $f^{-1}(f(A)) \neq A$. Indeed, if we consider the function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(n) := n^2$, then $f(\mathbb{N}) = \{0, 1, 4, 9, \dots\}$ but $f^{-1}(f(\mathbb{N})) = \mathbb{Z}$. Similarly, $B \supseteq f(f^{-1}(B))$ but $f(f^{-1}(B)) \neq B$. Indeed, for f given as above, $f^1(\mathbb{Z}) = \mathbb{Z}$, but $f(f^{-1}(\mathbb{Z})) = f(\mathbb{Z}) = \{0, 1, 4, 9, \dots\}$.

Definition 4.3.10

Let $f: X \rightarrow Y$ be a function.

1. We say f is **injective** or **one-to-one** if for every $y \in Y$, there exists one and only on $x \in X$ such that $f(x) = y$. In other words, for all $x, x_0 \in X$, $f(x) = f(x_0) \implies x = x_0$.
2. We say f is **surjective** or **onto** if for all $y \in Y$, there exists $x \in X$ such that $f(x) = y$. In other words $f(X) = Y$.
3. We say f is **bijective** if it is both injective and surjective.

5 September 11th: Compositions of Functions

5.1 Compositions of functions

Definition 5.1.1

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$. The **composition of f and g** , denoted $g \circ f: X \rightarrow Z$, is the function denoted by $g \circ f := g(f(x))$.

Example 5.1.2. For $f: \mathbb{N} \rightarrow \mathbb{N}$, $n \mapsto n + 1$ and $g: \mathbb{N} \rightarrow \mathbb{N}$, $n \mapsto n^2$, then $g \circ f := g(f(n)) = g(n + 1) = (n + 1)^2$, and $f \circ g := f(g(n)) = f(n^2) = n^2 + 1$.

Definition 5.1.3

We say the function $1_X: X \rightarrow X$, $x \mapsto x$ is the **identity function**.

Definition 5.1.4

Let $f: X \rightarrow Y$. Call $g: Y \rightarrow X$ the **right inverse of f** if $f \circ g = 1_Y$ and the **left inverse of f** if $g \circ f = 1_X$. Additionally g is the **inverse of f** if it is both the left and right inverse of f .

Remark 5.1.5. A composition of functions is associative, that is, if $f: X \rightarrow Y$, $g: Y \rightarrow Z$, $h: Z \rightarrow W$, then $h \circ (g \circ f) = (h \circ g) \circ f$.

Proposition 5.1.6

For $f: X \rightarrow Y$, if $g_l: Y \rightarrow X$ is a left inverse and $g_r: Y \rightarrow X$ is a right inverse, then $g_l \equiv g_r$ and thus is a two-sided inverse.

Proof. We can write $g_l = g_l \circ 1_Y = g_l \circ (f \circ g_r) = (g_l \circ f) \circ g_r = 1_X \circ g_r = g_r$. ■

Proposition 5.1.7

Let $f: X \rightarrow Y$, where $X \neq \emptyset$.

1. f admits an inverse if and only if f is bijective.
2. f admits a left inverse if and only if f is injective.
3. f admits a right inverse if and only if f is surjective.

Proposition 5.1.8

If $f: X \rightarrow Y$ has an inverse, then it is unique and we denote it by $f^{-1}: Y \rightarrow X$.

5.2 Binary Relations and Equivalence Relations

Definition 5.2.9

A **binary relation** on a set is a an arbitrary subset of $X \times X$.

Example 5.2.10. A

6 September 14th: Number Systems

6.1 Constructing the Rationals

We suppose that $\mathbb{N}$ is given by the set theory axioms. Then we can define $\mathbb{Z}$ as the union $\mathbb{N} \cup \{-n: n \in \mathbb{N}^+\}$ where $\mathbb{N}^+$ denotes the positive rational numbers. We can then define the order and multiplication and addition on $\mathbb{Z}$ the usual way though the equivalence classes $<$, $+$ and $\cdot$ on $\mathbb{N}$.

We can now define the set of rationals $\mathbb{Q}$. Begin by declaring $Q_0 := \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$. $Q_0 \neq \mathbb{Q}$, indeed $(1, 2) \neq (3, 6)$ while we know $\frac{1}{2} = \frac{3}{6}$. We define an equivalence relation on Q_0 by:

$$(a, b) (c, d) \iff ad = bc$$

One can easily check that is an equivalence relation, and we denote the $\sim$ -class of (a, b) by:

$$\frac{1}{2} := [(a, b)]$$

Now by definition, $\frac{1}{2} = \frac{3}{6}$. Denote the quotient by $\mathbb{Q}$, i.e. $\mathbb{Q} := \frac{\mathbb{Q}_0}{\sim}$. Define $+$ on $\mathbb{Q}$ as a function $+: \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ by:

$$\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd}$$

We now show that $+$ is well-defined, i.e. does not depend on the choices of representatives: let $\frac{a'}{b'} = \frac{a}{b}$ and $\frac{c'}{d'} = \frac{c}{d}$ and we show that:

$$\frac{a'd' + b'c'}{b'd'} = \frac{ad + bc}{bd}$$

The verification of this is an elementary exercise.

Similarly, define $\cdot$ and $\leq$.

Remark 6.1.1. Once this definition is understood, elements of $\mathbb{Q}$ may be thought of as special points on the real line as opposed to equivalence classes.

6.2 An Overview of Real Numbers

As we know from elementary math, each rational number admits a decimal representation. For instance $\frac{1}{3} = 0.3333\dots$, which is eventually periodic. We can define $\mathbb{R}$ as the set of all such decimal representations, i.e. as a set of sequences of digits $\{0, 1, \dots, 9\}$ with a period as one of the entries, and the first entry being the sign. However, this is not quite correct, say, $01.2000\dots$ should be the same as $1.2000\dots$, also $0.999\dots$ should be the same as $1.000\dots$, so we need to identify them using an equivalence relation. One can define $\leq$, $+$ and $\cdot$ in expected ways using addition and multiplication algorithms learnt in school.

6.3 The Field of Complex Numbers

Definition 6.3.2

We define the set of **complex numbers**, denoted $\mathbb{C}$, as the set $\{a + ib : a, b \in \mathbb{R}\}$ where $i = \sqrt{-1}$ and is known as the **imaginary solution** to $x^2 = -1$. Elements of $\mathbb{C}$ are called **complex numbers**.

We define $+$ and $\cdot$ on $\mathbb{C}$ using that $i^2 = -1$ and the usual distributivity laws:

1. $(a + ib) + (c + id) := (a + c) + i(b + d)$
2. $(a + ib) \cdot (c + id) := (ac - bd) + i(ad + bc)$

There is no natural order on $\mathbb{C}$ which respects $+$ and $\cdot$, so we don't define $\leq$ on $\mathbb{C}$.

Furthermore, for $z \in \mathbb{C}$ where $z := a + ib$, define the **real part** of z , denoted $\text{Re}(z) := a$ and the **imaginary part** of z , denoted $\text{Im}(z) := b$. We also write the absolute value of z as:

$$|z| := \sqrt{|\text{Re}(z)|^2 + |\text{Im}(z)|^2}$$

which is markedly similar to the notion of absolute values in Euclidean space. Hence we can interpret $\mathbb{C}$ geometrically as being akin to $\mathbb{R}^2$ where the x -coordinate is $\text{Re}(z)$ and the y -coordinate

is the imaginary part $\text{Im}(z)$, and the distance between points is the Euclidean distance:

$$|z - z'| := \sqrt{|\text{Re}(z) - \text{Re}(z')|^2 + |\text{Im}(z) - \text{Im}(z')|^2}$$

Definition 6.3.3

Let $z \in \mathbb{C}$. We say the **conjugate of** z , denoted $\bar{z} := \text{Re}(z) - \text{Im}(z)$ so for $z := a + ib$, $\bar{z} = a - ib$.

Observe that $z \cdot \bar{z} = |z|^2$.

Definition 6.3.4

Let $z \in \mathbb{C} \setminus \{0\}$. Define the **sign of** z , denoted $\text{sgn}(z)$, as:

$$\text{sgn}(z) := \frac{z}{|z|}$$

so $|\text{sgn}(z)| = 1$.

Observe, since $|\text{sgn}(z)| = 1$, we have $\text{sgn}(z) = \cos \theta + i \sin \theta =: e^{i\theta}$ for some $\theta \in \mathbb{R}$. This identity is called *Euler's equation*, and we define $\arg(z) := \{\theta \in \mathbb{R} : \text{sgn}(z) = e^{i\theta}\}$. Additionally, we can write $\arg(z) = \{\theta_n + 2\pi k : k \in \mathbb{Z}\}$ for some $\theta_n \in [0, 2\pi)$.

Thus it follows that we can write any $z \in \mathbb{C}$ in polar form as:

$$z = re^{i(\theta + 2\pi k)}$$

for all $k \in \mathbb{Z}$, where $r := |z|$ and $\theta \in [0, 2\pi)$.

Since $i = \sqrt{-1}$ and is a root to the equation $x^2 + 1 = 0$, we can define complex roots of polynomials that do not have real roots. This gives rise to the following theorem which we will revisit later:

Theorem 6.3.5 (Fundamental Theorem of Algebra)

Let $p(x) := a_0 + a_1x + \cdots + a_nx^n$ be a nonconstant polynomial over $\mathbb{C}$, i.e. $a_0, \dots, a_n \in \mathbb{C}$, $n \geq 1$ and $a_n \neq 0$. Then $p(x)$ has at least one root in $\mathbb{C}$.

Part II

Arithmetic of Integers

7 September 16th: Division

Definition 7.0.1

Let $a, b \in \mathbb{Z}$. We say that a **divides** b , denoted $a \mid b$, if there exists $q \in \mathbb{Z}$ such that $b = aq$.

Proposition 7.0.2

Let $a, b, c \in \mathbb{Z}$. The following statements hold:

1. $a \mid b \implies a \mid -b$.
2. $a \mid b \implies a \mid bc$.
3. $a \mid b$ and $a \mid c \implies a \mid (b \pm c)$.
4. $a \mid b$ and $b \mid c \implies a \mid c$.

The proofs of each of these statements are very easy by definition.

Theorem 7.0.3 (Quotient and Remainder Theorem)

Let $a, b \in \mathbb{Z}$ where $b \neq 0$. Then there exists $q, r \in \mathbb{Z}$ such that $a = bq + r$ and $0 \leq r < |b|$.

Proof. We must show existence and uniqueness of q and r in this proof. We begin by showing that q and r exist. Define:

$$\begin{aligned} S &:= \{r \in \mathbb{N} : r = a - bq, q \in \mathbb{Z}\} \\ &= \{a - bq : q \in \mathbb{Z}, a - bq \geq 0\} \end{aligned}$$

Then $S \subseteq \mathbb{N}$ and $S \neq \emptyset$. Indeed, if $a \geq 0$, then taking $q := 0$ gives $a = a - bq \in S$, and if $a < 0$ then take $q := \pm a$ with sign chosen such that $-bq \geq 0$, then $a - bq = a + |b||a| \geq a + |a| \geq 0$, whence $a + bq \in S$. By the well-ordering axiom of $\mathbb{N}$ there exists a least element $r \in S$. Since $r \in S$, we have $r = a - bq$ for some $q \in \mathbb{Z}$ and $r \geq 0$, so it remains to show that $r < |b|$. If we suppose by contradiction that $r \geq |b|$, then $r - |b| \geq 0$ and:

$$\begin{aligned} r - |b| &= a - bq - |b| \\ &= a - bq - \text{sgn}(b) \cdot b \\ &= a - b(q + \text{sgn}(b)) \in S \end{aligned}$$

which contradicts r being the least element in S .

We now show that r and q are unique. Suppose by contradiction, that $bq' + r = a = bq + r$ for some $q, q' \in \mathbb{Z}$ and $0 \leq r, r' < |b|$. Suppose without loss of generality that $r \geq r'$. Then $r - r' = bq' - bq = b(q' - q)$, so $b(q' - q) \geq 0$. If $q' - q = 0$ then $r - r' = 0$ and we are done. Otherwise, if $q' - q \neq 0$, then $|q' - q| \geq 1$ and so:

$$\begin{aligned} r &= r' + b(q' - q) \\ &= r' + |b||q' - q| \\ &\geq r' + |b| \\ &\geq |b| \end{aligned}$$

which contradicts that $0 \leq r' < |b|$. ■

Corrolary 7.0.4

Let $a, b \in \mathbb{Z}$ and $b \neq 0$. Then $b \mid a$ if and only if the remainder of a and b is 0.

Proof. For the forwards direction, assume $b \mid a$, then we have $a = bq$ for some $q \in \mathbb{Z}$, thus $a = bq + 0$, so by the uniqueness of such a representation, the remainder is 0. Conversely, suppose the remainder is zero, then for $a = bq + r$, if $r = 0$, $a = bq$ which immediately implies $b \mid a$. ■

8 September 18th: The GCD and The Euclidean Algorithm

8.1 The GCD

Definition 8.1.1

Let $a, b \in \mathbb{Z}$ such that a and b are nonzero. The **greatest common divisor** (often abbreviated **GCD**) of a and b , denoted $\gcd(a, b)$ is the largest positive integer d such that $d \mid a$ and $d \mid b$.

Theorem 8.1.2

Let $a, b \in \mathbb{Z}$ nonzero. Let $d := \gcd(a, b)$. Then every common divisor of a and b also divides d . Furthermore, there exists $u, v \in \mathbb{Z}$ such that:

$$d = ua + vb$$

Moreover, d is the smallest positive integer of the form $d = ua + vb$.

Proof. We first prove $d = ua + vb$. Let $S := \{ma + nb : m, n \in \mathbb{Z}, ma + nb \geq 1\}$. Note that $S \neq \emptyset$ because $a \operatorname{sgn}(a) + b \operatorname{sgn}(b) = |a| + |b| \geq 1$. By the well ordering property of $\mathbb{N}$, there is a least element $ua + vb \in S$. In particular, define $D := ua + vb \geq 1$. Since $d \mid a$ and $d \mid b$, $d \mid D$, thus $d \leq D$. To show $D \leq d$, it suffices to show that $D \mid a$ and $D \mid b$ by the definition of the GCD. Write $a = Dq + r$ for $r \in [0, D)$, then we write:

$$\begin{aligned} r &= a - Dq \\ &= a - uqa = vqb \\ &= (1 - uq)a - vqb \end{aligned}$$

But $r < D$ so if $r \neq 0$, then $r \in S$, which contradicts the choice of D . By the same argument, $D \mid b$ and $D \mid a$, so $D \leq d$ which implies $D = d$.

Now let $d' \mid a$ and $d' \mid b$ for some $d' \in \mathbb{Z}^+$. Then $d' \mid d$ since $d = ua + vb$. ■

Corrolary 8.1.3

For $a, b, c \in \mathbb{Z}$, $a \neq 0$, if $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.

Proof. We have that $1 = ua + vb$ for some $u, v \in \mathbb{Z}$, then $c = uac + vbc$ so $a \mid c$ since $a \mid uac$ and $a \mid vbc$. ■

8.2 The Euclidean Algorithm

How do we find the GCD of two integers a and b . One way is to brute force through all positive integers less than or equal to $\min(|a|, |b|)$, but this would take $\min(|a|, |b|)$ steps. A much more effective algorithm, which takes $\log_2 \min(|a|, |b|)$ steps, is known as the *Euclidean Algorithm* and was discovered over 2300 years ago.

Here is how we use the Euclidean algorithm to find the GCD of two positive integers a, b with $a \geq b$. We write:

$$\begin{aligned} a &= bq_0 + r_0, & 0 < r_0 < b \\ b &= r_0q_1 + r_1, & 0 < r_1 < r_0 \\ r_0 &= r_1q_2 + r_2, & 0 < r_2 < r_1 \\ &\vdots \\ r_{t-1} &= r_tq_{t+1} + r_{t+1} \end{aligned}$$

Let $t \in \mathbb{N}$ such that $r_{t+1} = 0$. Then $r_t = \gcd(a, b)$

Example 8.2.4. Find $\gcd(111, 51)$ using the Euclidean algorithm.

Solution. We write:

$$\begin{aligned} 111 &= 2 \cdot 51 + 9 \\ 51 &= 5 \cdot 9 + 6 \\ 9 &= 1 \cdot 6 + 3 \\ 6 &= 2 \cdot 3 + 0 \end{aligned}$$

Hence $\gcd(111, 51) = 3$. We can also use this algorithm to write 3 as a linear combination of 111 and 51 (in the form $3 = ua + vb$ using backwards substitution as follows:

$$\begin{aligned} 3 &= 9 - 6 \\ &= 9 - 51 + 9 \cdot 5 \\ &= 9 \cdot 6 - 51 \\ &= 6 \cdot 111 - 2 \cdot 51 \cdot 6 - 51 \\ &= 6 \cdot 111 - 11 \cdot 51 \end{aligned} \quad \checkmark$$

We can prove that the Euclidean algorithm returns $\gcd(a, b)$ by using recursion. Write:

$$\text{Euc}(a, b) := \begin{cases} b & \text{if } b \mid a \\ \text{Euc}(b, \text{rem}(a, b)) & \text{otherwise} \end{cases}$$

where $\text{rem}(a, b)$ is the remainder of a and b . It suffices to prove the following lemma:

Lemma 8.2.5

For $a, b \in \mathbb{N}^+$, with $a \geq b$, $\gcd(a, b) = \gcd(b, \text{rem}(a, b))$.

Proof. Let $r := \text{rem}(a, b)$, so $a = bq + r$ for some $q \in \mathbb{Z}$. Then $\gcd(b, r) \mid a$ because a is an integer combination of b and r . Since we also have $\gcd(b, r) \mid b$, we get $\gcd(b, r) \mid \gcd(a, b)$. Similarly, since $r = a - bq$, $\gcd(a, b) \mid r$. Since $\gcd(a, b) \mid b$ we finally get $\gcd(a, b) \mid \gcd(b, r)$. ■

9 Prime Numbers

9.1 Primes

Definition 9.1.1

An integer $p \in (\mathbb{Z} \setminus \{0, \pm 1\})$ is said to be **prime** if its only divisors are 1 and p . A **prime number** is a natural number which is prime, i.e. a number $p \geq 2$ such that its divisors are 1 and p .

Note that by definition, 1 is not a prime number.

9.2 The Sieve of Eratosthenes

To find prime numbers, we can use the *Sieve of Eratosthenes*, which is a slow but reliable algorithm to find all prime numbers from 2 to a natural number N .

Suppose, for sake of exposition, we want to find all the prime numbers between 2 and 50. We first write out all numbers between 2 and 50:

2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28
29	30	31	32	33	34	35	36	37
38	39	40	41	42	43	44	45	46
47	48	49	50					

We start by highlighting the smallest number—in this case it is 2. 2 is a prime number. We then cross out all multiples of 2, as they are not prime:

2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28
29	30	31	32	33	34	35	36	37
38	39	40	41	42	43	44	45	46
47	48	49	50					

Now, we move to highlight the next smallest number: 3. We now cross out all multiples of 3 (we

can skip numbers that have already been crossed out):

2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28
29	30	31	32	33	34	35	36	37
38	39	40	41	42	43	44	45	46
47	48	49	50					

The next smallest number, 4, has been crossed out (hence is not prime), so we move to highlight 5, which is prime. We now cross out all multiples of 5:

2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28
29	30	31	32	33	34	35	36	37
38	39	40	41	42	43	44	45	46
47	48	49	50					

Continuing this algorithm until all numbers have been either crossed out or highlighted, we get the prime numbers up to 50:

2	3	5	7	11	13	17	19	23	29	31	37	41	43	47
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9.3 Properties of Prime Numbers

Theorem 9.3.2 (Characterisation of Primes)

Let $p \geq 2$ be a natural number. p is prime if and only if for all $a, b \in \mathbb{Z}$, if $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. Assume that p is prime and that for all $a, b \in \mathbb{Z}$, $p \mid ab$. Assume $p \nmid a$, then $\gcd(a, p) = 1$ since p is prime. Thus, by Corollary 8.1.3, $p \mid b$. Conversely, suppose that for all $a, b \in \mathbb{Z}$, $p \mid ab$ implies $p \mid a$ and $p \mid b$. We suppose $p = st$ for some $s, t \in \mathbb{N}^+$ and show that $s = 1$ or $t = 1$. Since $p \mid st$, $p \mid s$, or $p \mid t$, say $p \mid s$. Then $s = ps'$ hence $p = st = s't$ so $s't = 1$, hence $s' = t = 1$. ■

Corollary 9.3.3

If $p \geq 2$ is prime and $p \mid a_1 a_2 \dots a_n$ for $a_1, \dots, a_n \in \mathbb{Z}$, then there exists an $i \leq n$ such that $p \mid a_i$.

Proof. Can be done trivially by induction of n using the Characterisation of Primes. ■

Theorem 9.3.4

Every natural number $n \geq 1$ is a product of primes.

Proof. Shown in homework. ■

Corrolary 9.3.5

Every natural number $n \geq 2$ is divisible by a prime.

Proof. TBD. ■

Corrolary 9.3.6

[Infinitude of Primes] There are infinitely many prime numbers.

Proof. Suppose by contradictoion that there are finitely many primes $p_1, \dots, p_k$ for some $k \geq 1$. Let $n := p_1 p_2 \dots p_k + 1$. Then there is a prime $q \mid n$ by the previous corrolary, so $q = p_i$ for some $i \in \{1, \dots, k\}$ since $p_1, \dots, p_k$ are all the primes. But then $1 = n - p_1 \dots p_k$ is divisible by $q = p_i$ since $q \mid a$ and $q \mid p_1 \dots p_k$, which is a contradiction. ■

Theorem 9.3.7 (Fundamental Theorem of Arithmetic)

Every nonzero integer n is a unique product of prime integers, i.e. $n = \sigma p_1 \dots p_k$, where $\sigma = \pm 1$ and $2 \leq q_1 \leq \dots \leq q_k$ are primes.

Proof. The existence of such a representation follows from Theorem 9.3.4. We only need to show the uniqueness of such a representation.

Let $\sigma p_1 \dots p_k = n = \sigma' q_1 \dots q_m$ where p_i and q_j are prime numbers ordered $2 \leq p_1 \leq \dots \leq p_k$ and $2 \leq q_1 \leq \dots \leq q_m$. The sign must be the same, i.e. $\sigma = \sigma'$, so assume without loss of generality that $n \geq 0$ and $p_1 \dots p_k = n = q_1 \dots q_m$. We prove that $k = m$ and $p_i = q_i$ for all $1 \leq i \leq k$. We can show this holds by strong induction on $n \geq 1$.

If $n = 1$, then $k = 0 = m$ and we are done. If we suppose $n \geq 2$, then $k, m \geq 1$ and without loss of generality suppose $p_1 \leq q_1$. Then $p_1 \mid q_1 \dots q_m$ so $p_i \mid q_j$ for some $1 \leq j \leq m$. But q_j is prime so $p_1 = q_j$. Since $q_1 \leq \dots \leq q_j$ and $p_1 \leq q_1$, we get $p_1 = q_1 = \dots = q_j$, so $n' := p_2 p_3 \dots p_k = q_2 q_3 \dots q_m$. By the induction hypothesis on n' , we get $k = m$ and $p_i = q_i$ for all $i \in [2, k]$, hence $k = m$ and $p_i = q_i$ for all $i \in [1, k]$, whence the representation is unique. ■